- *Advanced information extraction*: They can identify entities, relationships, and events from domain-specific texts, such as recognizing genes in biomedical literature or detecting legal clauses in contracts.
- *Text generation and summarization*: They can generate high-quality, domain-specific content and create accurate summaries of complex domain-specific texts.
- *Data-driven predictions and recommendations*: They can analyze domain-specific data for forecasting and providing recommendations, like predicting financial trends or suggesting personalized medical treatment plans.
- *Conversational agents and expert systems*: They can be incorporated into conversational agents or expert systems for domain-specific guidance, such as virtual tutors or legal chatbots.
- *Automated code generation and analysis*: In software engineering, they can generate or analyze code, identify bugs, or suggest improvements based on natural language descriptions.

In this section, we dive deep to review existing techniques for specializing LLMs in domain-specific tasks and discuss potential open challenges in detail. Due to the space limitation, we only provide a brief introduction of each domain and leave the complete discussion in the supplementary material.

Biomedicine. Language models are becoming increasingly useful in the field of biology, from fundamental biomedical research [91, 132] to clinical healthcare support [58, 105, 127]. At the fundamental biomedicine science level, LLMs can be trained on vast amounts of domain-specializing data (e.g., genomic and proteomic) to analyze and predict biological functions, disease mechanisms, and drug discovery. LLMs can also aid in predicting protein structures and interactions, which are critical for understanding cellular processes and designing new drugs. At the clinical healthcare support level, pre-trained or medical corpus fine-tuned LLMs can be used for the natural language processing of medical records to identify patterns, make diagnoses, and provide personalized treatment recommendations. Also, LLMs can assist in medical image analysis in a multi-modality learning way, such as identifying specific features in X-rays or MRI scans. Overall, LLMs offer tremendous potential for advancing biology research and improving healthcare outcomes.

Earth Science. Earth science is an interdisciplinary domain focused on examining the interactions between physical and human systems across diverse spatial and temporal scales. This field incorporates methods from Earth observation, information science, spatial analysis, complexity theory, and simulation modeling to investigate phenomena like climate change, land-use change, natural disasters, environmental development, and urbanization. Spatial information is vital to Earth science, and geographic information science tools are invaluable for a wide range of interdisciplinary studies involving spatial data. Large language models like ChatGPT can act as question-answer systems, assisting those interested in Earth Science to gain pertinent knowledge, such as recommending the optimal earth observation dataset for specific research purposes, offering code examples like Google Earth Engine code for processing satellite data, providing high-quality responses to environmental-related questions [188], developing innovative idea [9], generating climate scenario [9]. LLMs can also be tailored to various Earth Science-related downstream tasks through methods such as fine-tuning, few-shot, or even zero-shot learning.

Finance and Law. Specializing LLMs in the financial and legal domains requires careful adaptation to the distinctive characteristics of these fields. In the financial domain [71, 87, 169, 174], models need to comprehend complex financial terminologies, economic trends, and regulatory norms to accurately generate content like financial reports, investment analyses, or risk assessments. Meanwhile, the legal domain [15, 120, 151] demands understanding and generation of intricate legal language, comprehension of laws, legal codes, and court rulings, while maintaining absolute precision and a xxformal tone. For both domains, model specialization often involves fine-tuning with domain-specific datasets,

incorporating explicit domain knowledge, and optimizing for domain-specific objectives like compliance with regulations, accuracy of information, or effectiveness of advice. However, it's crucial to maintain an ethical guardrail for these models, given the high stakes nature of both financial and legal decisions. The specialized models also need to keep abreast of the evolving landscapes of these domains, adapting to changes in laws, regulations, or financial trends.

Human Computer Interaction and Software Engineering. Specializing LLMs in the domains of human-computer interaction (HCI) and software engineering requires a deep understanding of the terminologies, workflows, and conventions unique to these areas. In the HCI domain, an LLM may be specialized to understand and respond to user inputs more effectively, potentially improving the design and usability of interfaces by offering more natural and intuitive interaction paradigms. This involves training the model on diverse data, ranging from human conversational data to user interaction logs. On the other hand, in the software engineering domain, an LLM can be specialized to aid in tasks such as code generation, bug detection, code review, and documentation. This involves training the model on large codebases, issue trackers, documentation, and other software-related data. These specialized models can provide valuable assistance to developers, enhance the software development process, and potentially revolutionize the way we interact with computers. Despite the promising applications, several challenges remain, including the complexity of these domains, the need for accurate and up-to-date data, and the balance between specialized and general knowledge.

7 OPEN CHALLENGES AND FUTURE WORKS

7.1 Open Challenges

After having explored the current approaches for domain specialization of large language models, categorized as white box, grey box, and black box methods, it is essential to acknowledge that despite the significant progress in this field, there remain several open challenges. These challenges pervade all categories of models, irrespective of their accessibility or the specific techniques employed for specialization. As we strive to create LLMs that can effectively understand and generate domain-specific content, it is these challenges that will shape the future trajectory of research in this field. Let us delve into these open challenges to better comprehend the complexities of domain specialization and the areas where further research is required.

- *Domain Complexity*: Each domain has its unique intricacies and complexities, which could range from highly specialized vocabularies, and nuanced terminologies to complex knowledge structures. For instance, the legal or medical field employs language and terms that are extremely domain-specific and follow certain syntax and structure rules. This complexity extends to the relationships between different entities and concepts within the domain. Accurately understanding and modeling this intricate domain knowledge is a significant challenge for all types of models.
- *Balancing General and Domain Knowledge*: An LLM, while needing to understand the specificities of a particular domain, also has to maintain its general knowledge to provide contextually appropriate responses. If a model is overly specialized, it may perform exceptionally within the targeted domain but fail to understand or generate coherent responses to prompts outside of it. Conversely, retaining too much general knowledge may dilute the domain-specific responses. Striking this balance between general and domain knowledge is a complex task.
- *Explainability and Trust*: As LLMs become more sophisticated, their decision-making process also becomes more opaque, raising the challenge of explainability. It is crucial for users, especially in high-stakes domains like healthcare, law, or finance, to understand how the model arrived at a certain output. Achieving this transparency

can help build trust in the system. The challenge lies in the trade-off between model complexity and explainability, as increasing one often decreases the other.

- *Adapting to Domain Evolution*: Domains are not static; they evolve over time with the introduction of new terminologies, concepts, and trends. For example, the ongoing COVID-19 pandemic introduced a slew of new medical terms and concepts. Therefore, an LLM that is specialized for a certain domain must continuously adapt to these changes to stay relevant and effective. Designing models that can keep pace with the evolving landscape of their specialized domain is a challenging task.
- *Scalability*: Domain specialization often involves training or fine-tuning the LLM with domain-specific data, crafting specific prompts, or using other domain-specific resources. While this might be feasible for a few domains, scaling this process to cover a wide range of domains or to handle large, complex domains is a significant challenge. It involves not just computational resources but also the availability of domain-specific data and expertise. The challenge is to create efficient and effective methods for domain specialization that can be scaled to cover many different domains.

7.2 Future Works

As we chart the frontier of large language model specialization, it's not only crucial to build upon and refine the existing black-box, grey-box, and white-box methods, but also to anticipate and explore innovative, out-of-the-box techniques that have the potential to transcend these conventional approaches. Leveraging the rapid advancement of AI technologies and the increased understanding of LLMs, we can envision a future where novel techniques will emerge that push the boundaries of what's possible in domain specialization, providing enhanced performance, greater flexibility, and more efficient utilization of resources.

- *Hybrid Approaches*: This could involve combining multiple methods depending on the stage or specific needs. For example, a model could start with a black-box approach, using external resources to augment input prompts, then proceed to a grey-box method where gradients or loss values are used to further refine the prompts, and finally employ a white-box approach to fine-tune the model based on the learned strategies and feedback. This hybrid approach could provide a balance between resource requirement and model performance and might be especially effective when dealing with scarce domain-specific data.
- *Meta-Learning or AutoML Techniques*: AutoML or meta-learning strategies could be used to automate the process of selecting the best strategies for domain specialization. For instance, a meta-learning approach might learn a policy to select the best data for fine-tuning, the best prompt engineering techniques, or the best layers to fine-tune for a given domain, based on previous experience with similar domains. This could significantly reduce the resources and expertise needed for domain specialization, and could potentially lead to more effective and efficient methods.
- *Incorporating More Explicit World Knowledge*: Instead of relying solely on text-based pre-training, future LLMs might leverage structured knowledge sources, like knowledge graphs, to augment their understanding of the domain. This could involve techniques like graph neural networks or attention mechanisms that operate on graph-structured data. For instance, a medical LLM could incorporate knowledge from a medical ontology graph to better understand the relationships between various medical terms and concepts. This could lead to more accurate and informative outputs, especially in domains where explicit structured knowledge is available.